MOHAMMED SHABAZ S

Software & Embedded Engineer

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SUMMARY

Final-year ECE student (CGPA 8.38) targeting Telecom/5G engineering roles. Completed LTE and 5G study during a Nokia internship, covering network architecture, interfaces, call flows, mobility, RRM, O-RAN and troubleshooting. Brings software and embedded strengths from application work and microcontroller projects.

TECHNICAL SKILLS

Programming: C, C++, Python, SQL (Intermediate); Java (currently learning)
Telecom / 5G: LTE, EPC, interfaces, call flows, QoS, RRM, LTE-Advanced; 5G, 3GPP Release 18, gNodeB, RAN, EN-DC, VoNR, NSA/SA mobility, Xn, O-RAN, troubleshooting (Nokia internship study)
Embedded: Embedded C, Microcontrollers (Intermediate); ARM Cortex-M, FSM, non-blocking design, GPIO, I2C, ADC, timers, sensor interfacing
Tools & Foundations: Git, GitHub; Testing/Debugging (Basic); DSA, DBMS, OS, Computer Networks (Basic); Robot Framework (currently learning)

EXPERIENCE

Student Intern 16 September 2026 – Present
NOKIA SOLUTIONS AND NETWORKS INDIA PRIVATE LIMITED *Bangalore*

- Completed LTE and 5G study as part of the internship, covering mobile-network architecture, call flows, mobility, radio-resource management, O-RAN and troubleshooting.

AI Research Intern – Deep Learning & Model Development 23 March 2026 – 23 August 2026
iHelp Robotics Private Limited · Project: MITRA *Remote*

- Contributed Android app-side software and testing for MITRA, an iHelp company product. Work focused on live-stream reliability, app status handling and voice-command robustness; backend and model work were handled by a separate team.

Student Intern 1–30 June 2026
IEEE EMBS Pune Section *Remote*

PROJECTS

Automotive Body Control Module GitHub
ESP32, Arduino/C++, Finite-State Machine, GPIO, I2C OLED

- Programmed an ESP32 body-control module with a 3-state (OFF/ACC/ON) ignition FSM driving 6+ vehicle functions: turn indicators, synchronized hazard, brake logic, buzzer feedback and an OLED dashboard.
- Architected a non-blocking millis()-based scheduler (zero delay() in the main loop) with 30 ms brake debouncing, throttled 10 Hz I2C OLED refresh, a clean GPIO abstraction and edge-triggered serial logging for deterministic real-time behavior.

Smart Wellness Desk Assistant GitHub
STM32 Nucleo-L476RG (ARM Cortex-M4), STM32 HAL, STM32CubeIDE, Embedded C · College team project

- Interfaced ultrasonic presence and 12-bit ADC temperature sensors, an I2C SSD1306 OLED and timer-driven buzzer alerts on an STM32 Nucleo-L476RG (ARM Cortex-M4) using STM32 HAL, coordinated by a non-blocking finite-state machine; wired, tested and debugged in STM32CubeIDE as part of a college team.

VisionPay — Real-Time Indian Currency Detection GitHub
Python, OpenCV, TensorFlow Lite, MobileNetV2, Text-to-Speech

- Created an offline, real-time tool that recognizes Indian currency from a live webcam and speaks the result aloud for visually impaired users.
- Collected ~400 images per denomination across 6 classes (Rs.10–Rs.500) and trained a MobileNetV2 classifier converted to TensorFlow Lite, reporting ~93% validation accuracy.
- Strengthened real-time reliability with 4 mechanisms: confidence/margin gating, temporal smoothing, a background class, and an auto-count mode.

EDUCATION

B.E. Electronics & Communication Engineering Expected 2027
Ballari Institute of Technology and Management (BITM), Ballari *Final Year · CGPA 8.38*

LEADERSHIP & ACTIVITIES

- Vice-Chair, IEEE CAS Society, IEEE Student Branch BITM (previously Treasurer); Treasurer, BITM Robotics Club.
- Google Student Ambassador (2025); Selected Participant, IEEE SPACE 2026 B.Tech Initiative; participant, Techzone Nationals 2K25 Hardware Hackathon; Embedded AI workshop attendee.

CERTIFICATIONS

Embedded Systems — Internshala Trainings (2025) · Programming in C and C++ with AI · SQL for Data Analytics with AI — Internshala Trainings · Google Cloud Generative AI — SmartBridge/SmartInternz (2025) · Python Programming — EISystems (2024)